

ELOQUENT CLEF shared tasks for evaluation of generative language model quality

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Abstract. ELOQUENT is a set of shared tasks for evaluating the quality and usefulness of generative language models. ELOQUENT aims to bring together some high-level quality criteria, grounded in experiences from deploying models in real-life tasks, and to formulate tests for those criteria, preferably implemented to require minimal human assessment effort and in a multilingual setting. The selected tasks for this first year of ELOQUENT are (1) probing a language model for topical competence; (2) assessing the ability of models to generate and detect hallucinations; (3) assessing the robustness of a model output given variation in the input prompts; and (4) establishing the possibility to distinguish human-generated text from machine-generated text.

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Generative language models (“LLMs”) have become a foundational technology for both research and information service development in the past year, and have prompted great public interest through some quite innovative application scenarios.

A generative language model as a component in an information service can handle a broad variety of input data robustly and elegantly, and can provide appropriately creative generated output to fit the varied application situations and the preferences of a diverse user population. An information service with a generative language model can be built to provide a flexible low threshold conversational interface for its users: there is considerable interest to put generative language models to use in productive practical applications, in industry, NGOs, and public sector, that cover one or multiple languages.

Deploying generative models, or selecting among the several recent models with various generative capacities, is challenged by run-time costs, training costs, and to a great extent by concerns about the quality, predictability, relevance, appropriateness, and reliability of the output from the generative component. A language model can provide a system with fluently generated output but without guarantees of relevance, veracity, or consistency. In research contexts, quality assurance, evaluation, and benchmarking of large language models is performed

using tests that focus on fluency, and details about language understanding and some aspects of reasoning using language. There is a large number of tasks available for this purpose. However, many factors that are crucial for application development and deployment to third parties, such as veracity and robustness, are not addressed systematically through existing benchmarks. In addition, the current benchmarks are primarily Anglo-centric, which raises questions about the quality of the LLM output produced in other languages, or even in a multilingual setting.

The ELOQUENT evaluation lab will experiment with new evaluation methods for generative language models to meet some of the challenges in the path from laboratory to application. We will formulate a series of tasks to be developed iteratively over several years, each year drawing from experiences from the previous year. We recognise that organisations, whether commercial or not, will likely be unable and unwilling to expend great human effort on testing, benchmarking, sampling, and validation to perform quality assurance on a language module which oftentimes is a subsidiary component of a larger envisioned service [8]. We intend to develop tasks which do not rely on fixed gold standards and which involve minimal human effort expended on assessment, instead leveraging the intrinsic qualities of generative AI for test development and scoring.

The outcome of this lab is not only to score the performance of submitted models and to provide benchmarks for future reuse but to establish useful top-level quality criteria, which existing and new tests will contribute to assessing, and to more generally point out future directions for evaluation of generative and learning models [9]. Such top-level quality criteria include the directly linguistic notions of utterance-level correctness, clarity and fluency; of session-level discourse and conversational competence and awareness of social norms and conventions; of relevance, truthfulness, and topical quality, and of resilience and consistency in face of varying input. These and further quality criteria are partially covered by existing test suites. We intend to address some of the identified gaps in this lab.

The tasks are intended to involve comparatively little human effort, and instead leverage the capacity of the current generation of AI models to generate, process, and assess human language input. The organisers wish to encourage efforts in multilingual research and as hope to see participants work with models in many languages. Addressing the challenges of accommodating a variety of languages and cultural areas in a common evaluation framework is one of the intended focus topics of the workshop.

Task 1: Topical competence

A generative language model in practical application will in most envisioned use cases be expected to stay within topical boundaries, to generate material restricted to the domain it is employed to work within, and to have competence in the terminology and conventions of that domain. Examples of relevant topical domains could be business domains, such as finance [19] or healthcare [15], or

even recreational activities such as sailing or basketball, ranging to differences in how a topic is treated differently across linguistic and cultural areas or in specific demographic groups. We do not intend to constrain the type of topic to be included and will in this task experiment with a wide range of topics and discourse types, welcoming suggestions and proposals from task participants.

The discourse control structures and domain knowledge for generative language models are in today’s architectures implicit and learned via the acquisition of linguistic competence, but increasing interest has been shown towards a separate knowledge representation for domain information coupled to the input and output capacities of a language model.⁴ The generality of language models, and their ability to work across domains with little or no fine-tuning is part of their appeal, and verification of their capacity for a specific domain under consideration is usually done as part of a fine-tuning process. The more general question, given a foundation model, whether it already is competent for some application domain of interest is not addressed in general terms.

This task will both answer to the need for verifying a model’s understanding of an application domain of interest and will also in future iterations probe the capacity of a model to communicate with another model. This first year, the task will be for a model to generate and score topical tests, applied between participating models in a round-robin fashion. We expect this task to develop during the next years, with coming years including more challenges related to conversational competence with two or more models set up to engage in a conversation to solve some task which will involve making use of both topical and conversational competence.

Task 2: HalluciGen Detection

Detecting hallucinated or factually incorrect information may be difficult for humans. LLMs should therefore be rigorously tested for their ability to generate accurate output prior to deployment. One possible approach for detecting hallucinated content, that we will explore in this lab, is the use of LLMs to evaluate LLM output.

Previous work has framed the problem as one of cross-model evaluation where one model evaluates another’s output [10,14], or self-evaluation where the model evaluates its own output [10,12]. Inspired by the *Build it, break it: Language Edition* task [6], we divide the task into a builders subtask which focuses on the development of model-evaluators, and a breakers subtask which focuses on the development of challenging evaluation datasets.

The focus of Year 1 will be on the *builder task*. Participants will develop multilingual and monolingual hallucination-aware models (*evaluators*) that are able to both detect and generate hallucinated content. Given a pair of submitted models, A and B, and within a conversational setting, model A will be prompted to generate two versions of the same input containing either the presence/absence

⁴ <https://wsl.iitb.ac.in/cikm-2023-workshop-on-enterprise-knowledge-graphs-using-large-language-models/>

of hallucinated content, and model B will be asked to detect whether the output contains hallucinated content. We will leverage pre-existing manually-compiled datasets, e.g. data from the SHROOM task on *Hallucinations and Related Observable Overgeneration Mistakes* at SemEval 2024, to validate the ability of the models to generate and detect hallucinated content. For the generation step, we will measure the semantic similarity between model output and a human-authored gold standard, and for the detection step we will validate performance using a held-out test set, possibly machine-generated. This task will assess both model awareness of hallucination and the viability of cross-model evaluation.

The focus of Year 2 will be on the *breaker task*. Participants will develop novel evaluation datasets/benchmarks designed to *break* the hallucination detection capabilities of one or more baseline *evaluator* models. The evaluator model(s) could, for example, be the best-performing model(s) from Year 1. The aim of the task is to provide challenging examples, of human or near-human quality, that expose the strengths and weaknesses of the models in a monolingual or multilingual setting. This task is inspired by the WMT challenge set subtask [7] and the development of the Adversarial NLI dataset [13]. The focus will be on automated methods for dataset construction, for example leveraging pre-existing datasets and applying techniques such as data augmentation (e.g. using LLMs). This task will explore the use of automated methods for generating high-quality evaluation benchmarks.

Task 3: Robustness

Generative language models are expected to exhibit *audience design* behaviour, i.e. to fit their output to the preceding input [2]. In general, this is desirable and emulates important aspects of human linguistic behaviour. However, if this variation extends to content-related aspects of the output, this may have the unfortunate effect of systematically generating different material depending on user group, if e.g. the system is sensitive to dialectal, sociolectal, cross-cultural, or otherwise significant variability in linguistic items in the input.

This task will assess the how elicited responses from generative language models vary by input prompt variation. The task will organise a set of input prompts that vary along some dimensions of style and form, and the resulting outputs from participating systems will be scored for some observable content-related variables of interest. Robustness has been identified as a quality criterion when models have positional biases in responses to multiple choice questions [20] and in the face of adversarial attacks [18,11,1]. We intend to explore and extend this notion further in this task.

This task will, besides assessing the quality of participating models, establish quantitative measures for the range of elicited responses from the generative language models and for the robustness vs. volatility of generative language models with respect to variation in input.

Task 4: Voight-Kampff

Recent advances in LLMs have made it possible to automatically generate content for websites, news articles, social media, etc. Labelling AI-generated content, in order to tackle disinformation, is something that the EU has recently urged technology companies to do.⁵

This task is designed to explore whether automatically-generated text can be distinguished from human-authored text. Detecting automatically generated text, with the increased quality of generative AI, is becoming a task quite similar to human authorship verification. This task will be organised in collaboration with the PAN lab at CLEF, which allows us to build on years of experience from previous shared tasks organised by the PAN lab on authorship verification and closely related tasks. Recent editions of PAN have studied authorship verification from a *cross-domain* [17,5,4] and *cross-discourse type verification* perspective [16].

The Voight-Kampff task will be split into two subtasks, in the builder-breaker style, similar to the HalluciGen task described above. In the *builder* task PAN lab participants will build systems to discriminate between automatically-generated and human-authored texts [3]. The systems will be evaluated using the dataset(s) constructed by the *breakers*.

In the *breaker* task ELOQUENT lab participants will focus on constructing an evaluation dataset designed to challenge the discriminative capabilities of the *breaker* systems. A human-authored test set representing different styles and genres will be selected from existing texts, and the automated set will be generated by first automatically summarising the human-authored texts, and then instructing an LLM to generate a text of equal length to the original based on the summary. Participants may employ different strategies to evade detection by the *breaker* systems, for example designing LLMs to output text in a similar in style to the human-authored texts, or to inject human-like errors in the texts.

Invitation to participate

Those interested in participating in the ELOQUENT evaluation lab are encouraged to complete the online sign-up form. Participants will be invited to join the participant Google Group where we will discuss the formulation of the tasks in more detail.

References

1. Altinisik, E., Sajjad, H., Sencar, H.T., Messaoud, S., Chawla, S.: Impact of adversarial training on robustness and generalizability of language models (2023)
2. Bell, A.: Language style as audience design. *Language in society* **13**(2) (1984)

⁵ <https://www.theguardian.com/technology/2023/jun/05/google-and-facebook-urged-by-eu-to-label-ai-generated-content>

3. Bevendorff, J., Bonet Casals, X., Chulvi, B., Dementieva, D., Elnagar, A., Fröbe, M., Korenčić, D., Mayerl, M., Mukherjee, A., Panchenko, A., Potthast, M., Rangel, F., Rosso, P., Smirnova, A., Stamatatos, E., Stein, B., Taulé, M., Ustalov, D., Wiegmann, M., Zangerle, E.: Overview of PAN 2024: Multi-Author Writing Style Analysis, Multilingual Text Detoxification, Oppositional Thinking Analysis, and Generative AI Authorship Verification. In: *Advances in Information Retrieval: 46th European Conference on IR Research (ECIR) (2024)*
4. Bevendorff, J., Chulvi, B., la Peña Sarracén, G.L.D., Kestemont, M., Manjavacas, E., Markov, I., Mayerl, M., Potthast, M., Rangel, F., Rosso, P., Stamatatos, E., Stein, B., Wiegmann, M., Wolska, M., Zangerle, E.: Overview of PAN 2021: Authorship verification, profiling hate speech spreaders on twitter, and style change detection. In: *Experimental IR Meets Multilinguality, Multimodality, and Interaction - 12th International Conference of the CLEF Association*. vol. 12880 (2021)
5. Bevendorff, J., Ghanem, B., Giachanou, A., Kestemont, M., Manjavacas, E., Markov, I., Mayerl, M., Potthast, M., Pardo, F.M.R., Rosso, P., Specht, G., Stamatatos, E., Stein, B., Wiegmann, M., Zangerle, E.: Overview of PAN 2020: Authorship verification, celebrity profiling, profiling fake news spreaders on twitter, and style change detection. In: *Experimental IR Meets Multilinguality, Multimodality, and Interaction - 11th International Conference of the CLEF Association*. vol. 12260 (2020)
6. Ettinger, A., Rao, S., Daumé III, H., Bender, E.M.: Towards linguistically generalizable NLP systems: A workshop and shared task. In: *Proceedings of the First Workshop on Building Linguistically Generalizable NLP Systems*. pp. 1–10. Association for Computational Linguistics, Copenhagen, Denmark (Sep 2017)
7. Freitag, M., Rei, R., Mathur, N., Lo, C.k., Stewart, C., Avramidis, E., Kocmi, T., Foster, G., Lavie, A., Martins, A.F.T.: Results of WMT22 metrics shared task: Stop using BLEU – neural metrics are better and more robust. In: *Proceedings of the Seventh Conference on Machine Translation (WMT)*. Association for Computational Linguistics, Abu Dhabi (2022)
8. Karlgren, J.: Adopting systematic evaluation benchmarks in operational settings. *Information Retrieval Evaluation in a Changing World: Lessons Learned from 20 Years of CLEF* (2019)
9. Karlgren, J., Callin, J., Collins-Thompson, K., Gyllensten, A.C., Ekgren, A., Jurgens, D., Korhonen, A., Olsson, F., Sahlgren, M., Schütze, H.: Evaluating learning language representations. In: *Experimental IR Meets Multilinguality, Multimodality, and Interaction: 6th International Conference of the CLEF Association, CLEF’15, Toulouse, France, September 8-11, 2015, Proceedings 6*. Springer (2015)
10. Manakul, P., Liusie, A., Gales, M.J.F.: Selfcheckgpt: Zero-resource black-box hallucination detection for generative large language models (2023)
11. Moradi, M., Samwald, M.: Evaluating the robustness of neural language models to input perturbations (2021)
12. Mündler, N., He, J., Jenko, S., Vechev, M.: Self-contradictory hallucinations of large language models: Evaluation, detection and mitigation (2023)
13. Nie, Y., Williams, A., Dinan, E., Bansal, M., Weston, J., Kiela, D.: Adversarial NLI: A new benchmark for natural language understanding. In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. pp. 4885–4901. Association for Computational Linguistics, Online (2020)
14. Saunders, W., Yeh, C., Wu, J., Bills, S., Ouyang, L., Ward, J., Leike, J.: Self-critiquing models for assisting human evaluators (2022)

15. Singhal, K., Azizi, S., Tu, T., Mahdavi, S.S., Wei, J., Chung, H.W., Scales, N., Tanwani, A., Cole-Lewis, H., Pfohl, S., Payne, P., Seneviratne, M., Gamble, P., Kelly, C., Babiker, A., Schärli, N., Chowdhery, A., Mansfield, P., Demner-Fushman, D., Agüera y Arcas, B., Webster, D., Corrado, G.S., Matias, Y., Chou, K., Gottweis, J., Tomasev, N., Liu, Y., Rajkomar, A., Barral, J., Sennott, C., Karthikesalingam, A., Natarajan, V.: Large language models encode clinical knowledge. *Nature* **620**(7972), 172–180 (2023). <https://doi.org/10.1038/s41586-023-06291-2>, <https://doi.org/10.1038/s41586-023-06291-2>
16. Stamatatos, E., Kestemont, M., Kredens, K., Pezik, P., Heini, A., Bevendorff, J., Stein, B., Potthast, M.: Overview of the Authorship Verification Task at PAN 2022. In: Faggioli, G., Ferro, N., Hanbury, A., Potthast, M. (eds.) *CLEF 2022 Labs and Workshops, Notebook Papers*. CEUR-WS.org (Sep 2022)
17. Stamatatos, E., Potthast, M., Pardo, F.M.R., Rosso, P., Stein, B.: Overview of the PAN/CLEF 2015 evaluation lab. In: *Experimental IR Meets Multilinguality, Multimodality, and Interaction*. vol. 9283 (2015)
18. Wang, B., Wang, S., Cheng, Y., Gan, Z., Jia, R., Li, B., Liu, J.: Infobert: Improving robustness of language models from an information theoretic perspective. *arXiv preprint arXiv:2309.02329* (2021)
19. Wu, S., Irsoy, O., Lu, S., Dabrowski, V., Dredze, M., Gehrmann, S., Kambadur, P., Rosenberg, D., Mann, G.: Bloomberggpt: A large language model for finance (2023)
20. Zheng, C., Zhou, H., Meng, F., Zhou, J., Huang, M.: Large language models are not robust multiple choice selectors. *arXiv preprint arXiv:2309.03882* (2023)